# Charlottetown and Stratford Urban Growth

**Group 2 Project Proposal | Option 4**

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### Research Question

How many hectares of agricultural and vegetated land were converted to urban/peri-urban land in Charlottetown and Stratford between 1935 and 2020? How much does classification accuracy decrease when colour is removed from the 2020 imagery, particularly when separating agricultural and vegetated land?

### Data and Time Periods

We will use five years: **1935, 1958, 1968, 2000, and 2020**. The historical images are single-band grayscale; only 2020 provides RGB colour. We will retain 1958 and mask its missing areas. All no-data pixels, including water cutouts in 1935 and 1968, will be excluded from training, evaluation, and change calculations. Imagery and labels will be aligned to a common projected grid before comparison.

### Methods

**Classification:** We will map urban/peri-urban, agricultural, vegetated, and water classes where valid imagery supports them. Larger patches will provide surrounding context to distinguish lawns and parks within developed areas from pasture.

**Baseline and U-Net:** We will train a Random Forest using image intensity and GLCM texture features, then assess whether fine-tuning a compact U-Net improves segmentation. U-Net feasibility will depend on label quality, validation performance, and computing time.

**Colour experiment:** We will compare RGB and grayscale U-Nets using identical 2020 patches, labels, spatial splits, and matched training settings. Changes in mean IoU and agricultural/vegetated F1 will measure the effect of removing colour within 2020.

### Evaluation Plan

We will manually label 40 patches of 250 $\times$ 250 m per year across both sites combined, giving 200 patches covering suburban fringes, agricultural land, and vegetated areas.

We will use spatially separate training (60%), validation (20%), and test (20%) sets, keeping all dates from the same location in the same set.

Both models will be evaluated using mean IoU and per-class F1. Comparing 2020 RGB and grayscale results will measure the effect of removing colour. Transition matrices will estimate hectares converted from agricultural or vegetated land to urban land, using only pixels valid in both comparison years.

### Feasibility and Risks

We will assess feasibility through early labeling and model tests on Google Colab.

- **Biggest Risk:** Too few labels may cause lawn/pasture confusion in grayscale imagery. Larger patches will provide surrounding context.
- **Fallback Plan:** If U-Net is unreliable, we will use Random Forest for the main analysis and matched RGB/grayscale Random Forest models for the colour comparison.
- **First Test:** We will label a small set of representative patches containing lawns, pasture, and vegetated land. We will record labeling time and test the Random Forest to assess the labeling workload and identify classification errors before training U-Net